Master Multimedia Networking: Content Distribution Networks, MN905

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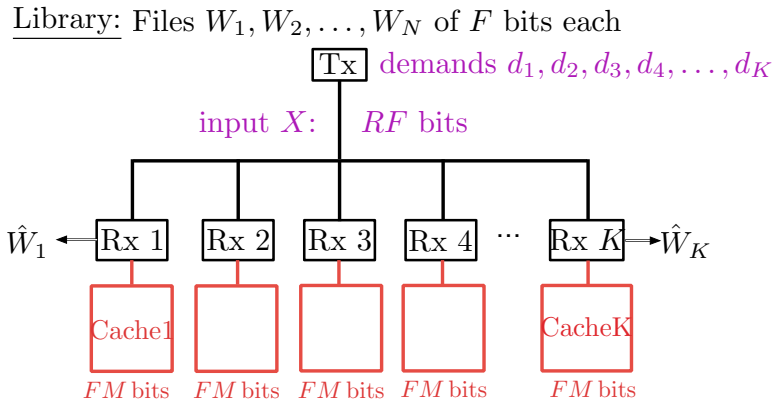
Instructions: You have 30 min to solve the exam.
Try to look at all the questions in all the three problems. You might find some of them easier than others. Moreover, sometimes you might be able to answer for example question 3.) even if you could not answer the previous question 2.). Notice also that at any given time in the exam you can use all the statements that you were asked to prove in previous parts of the exam, even if you were not able to prove them.
Make sure you justify your answers! Your answers are graded also according to your justifications.

Problem 1 *4 points*

Consider the following network with a single server and K receivers, where each receiver has a cache memory of size MF bits. The transmitter has a library with N files $W_1, \dots, W_N$, each consisting of



F random bits, and can store an arbitrary function of these messages in each of the cache memories of receivers $1, \dots, K$. Provided that it satisfies the memory constraint. Just prior to the delivery phase the receivers reveal their demands $d_1, d_2, \dots, d_K \in \{1, \dots, N\}$ to all terminals. Depending on these demands, during the delivery phase, the transmitter sends an input X consisting of RF bits over a noise-free bit-pipe to each and every receiver. The goal of the communication is that each Receiver $k \in \{1, \dots, K\}$ can reconstruct file W_{d_k} from X and from the content pre-stored in its cache memory.
Assume now that $K = 6$ and $N = 12$.

1. Is the rate-memory pair $M = 8$ and $R = 2/5$ achievable? *(1pts)*
2. Find the best *upper bound* on the rate-memory tradeoff $R^*(M)$ that we have seen in class for $M = 1$. *(1.5pts)*
3. Find the best *lower bound* on the rate-memory tradeoff $R^*(M)$ that we have seen in class for $M = 1$. *(1.5pts)*

Problem 2 *4 points*

Consider the same setup as in Problem 1, except that now each receiver demands multiple files. Assume that each receiver demands 3 files and that $N > 3K$. So, instead of sending a single demand to the other terminals, before the delivery phase, it sends a quadruple $\mathbf{d}_k = (d_{k,1}, d_{k,2}, d_{k,3}) \in \{1, \ldots, N\}^3$. The goal of the communication is that each Receiver k can reconstruct files $W_{d_{k,1}}, W_{d_{k,2}}$, $W_{d_{k,3}}$ from X and from the contents pre-stored in its cache memory.

1. Find an *upper bound* on the minimum delivery rate R in function of the cache size M so that each receiver can reconstruct its 3 demanded files. *(2pts)*

2. Find a *lower bound* on the minimum delivery rate R in function of the cache size M so that each receiver can reconstruct its 3 demanded files. *(2pts)*